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3.3 PRACTICAL EXAMPLES WITH COILS AND TRANSFORMERS

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On the figure 2.6b coils, along with the capacitor, form two filters for conducting the currents to the speakers. The coil and capacitor C on figure 2.6c form a parallel oscillatory circuit for “amplifying” a particular radio signal, while rejecting all other frequencies.

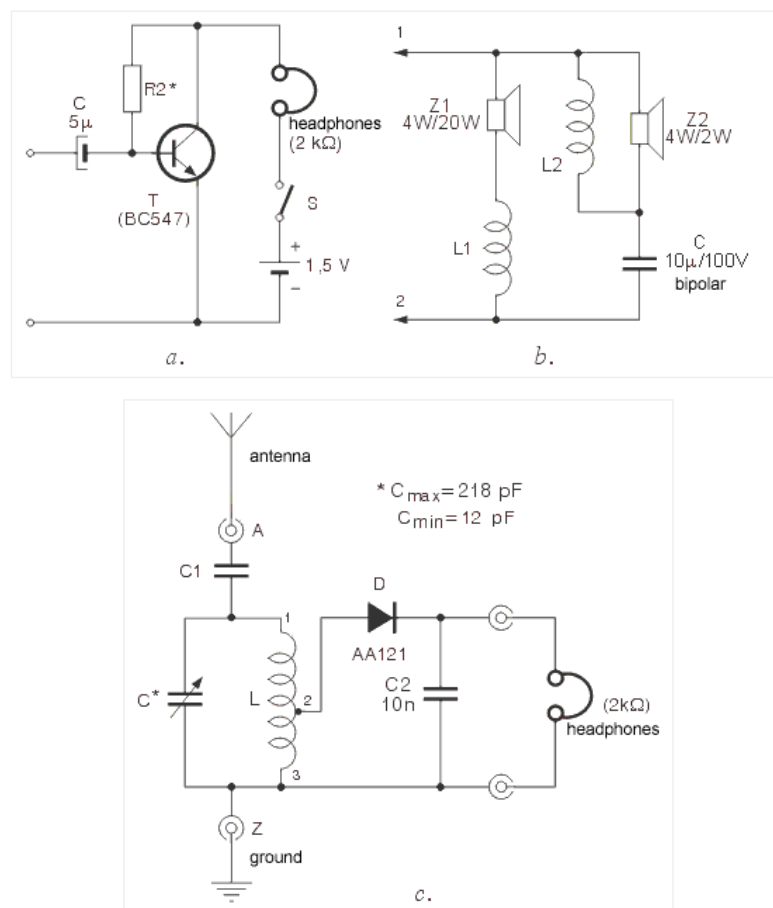


Fig. 2.6: a. Amplifier with headphones, b. Band-switch, c. Detector radio-receiver

The most obvious application for a transformer is in a power supply. A typical transformer is shown in figure 3.8 and is used for converting 220V to 24V.

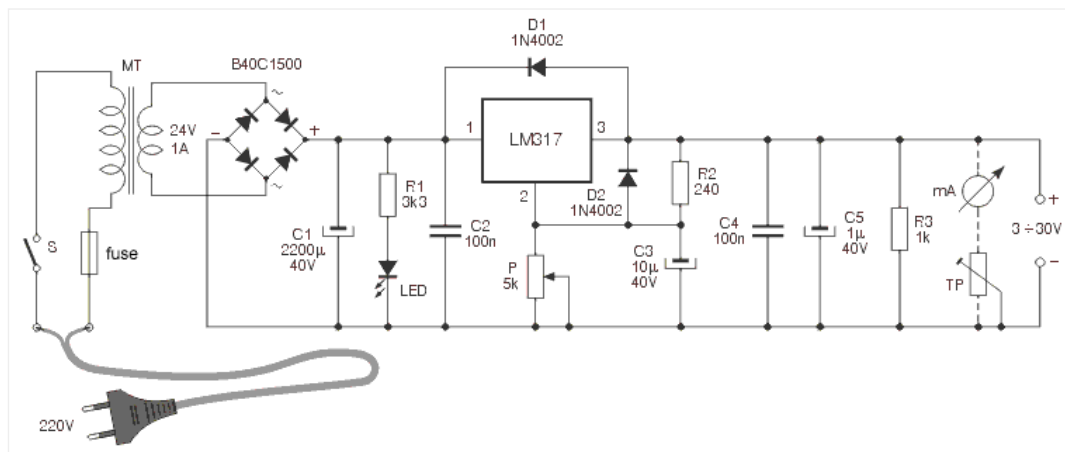


Fig. 3.8: Stabilized converter with circuit LM317

Output DC voltage can be adjusted via a linear potentiometer P, in 3~30V range.

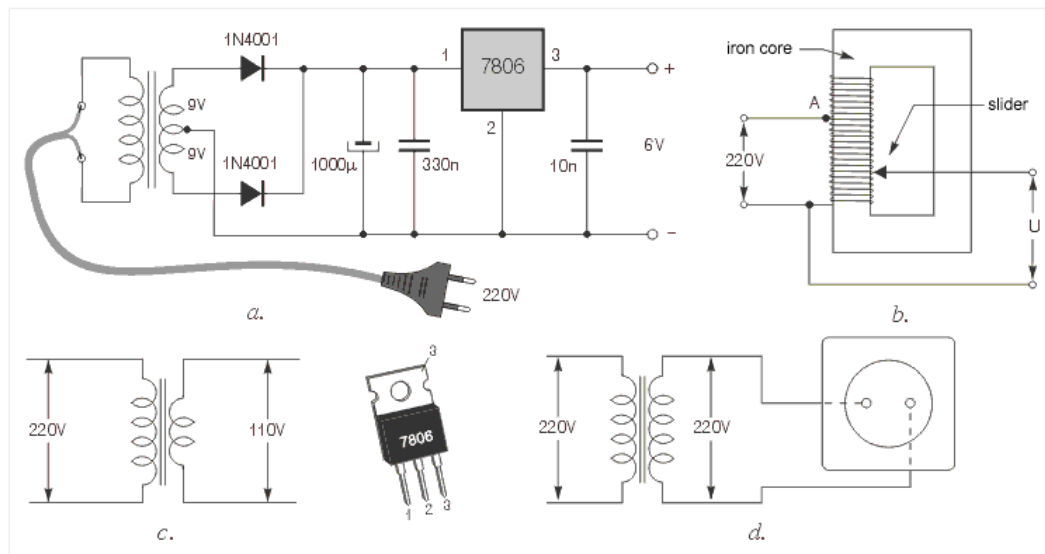


Fig. 3.9: a. Stabilized converter with regulator 7806, b. auto-transformer, c. transformer for devices working at 110V, d. isolating transformer

Figure 3.9a shows a simple power supply, using a transformer with a centre-tap on the secondary winding. This makes possible the use two diodes instead of the bridge in figure 3.8.

Special types of transformers, mainly used in laboratories, are auto-transformers. The diagram for an auto-transformer is shown in figure 3.9b. It features only one winding, wound on an iron core. Voltage is taken from the transformer via a slider. When the slider is in its lowest position, voltage equals zero. Moving the slider upwards increases the voltage U , to 220V. Further moving the slider increases the voltage U above 220V.

The transformer in figure 3.9c converts 220v to 110v and is used for supplying devices designed to work on 110V.

As a final example, figure 3.9d represents an isolating transformer. This transformer features the same number of turns on primary and secondary windings. Secondary voltage is the same as the primary, 220V, but is completely isolated from the “mains,” minimizing the risks of electrical shock. As a result, a person can stand on a wet floor and touch any part of the secondary without risk, which is not the case with the normal power outlet.



